Graphics

Higher · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain how graphics are used to generate, explore, refine and communicate ideas
- ☐ choose a suitable graphic for a purpose: orthographic, sectional, exploded, pictorial, working drawing, render
- ☐ describe the advantages of manual and computer-generated graphics
- ☐ describe what a working drawing shows

I DO — Worked example

Read how a full-mark answer is built

1 Outline two pieces of information shown on a working drawing. (2 marks)

Full-mark answer each ✓ is one mark

All the dimensions needed to make each part ✓. Orthographic views of every component with the material each is made from ✓.

Why answers lose marks

"It shows what the product looks like and how it works."

0 marks: both responses were credited 0 in SQA marking.

WE DO — Guided practice

Use the starters and word bank to help you

A designer presents a speaker as a computer-generated render.

2 Describe two benefits of using computer-generated graphics. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

file · proportion · texture · realistically · CAM · aesthetics · client · colour

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which graphic best shows the inside of a product?
☐ Exploded view ☐ Sectional view ☐ One-point perspective ☐ Mood board

b) Why would a designer show a client a rendered 3D illustration rather than an orthographic drawing?
☐ It includes all dimensions ☐ Non-experts find it easier to understand ☐ It is quicker than sketching ☐ It is needed for CNC

c) Which is an advantage of CAD over manual drawing?
☐ No training needed ☐ Changes can be made without redrawing ☐ Cheaper equipment ☐ Better at very rough early ideas

d) Which graphic communicates dimensions most clearly?
☐ Rendered illustration ☐ Orthographic drawing ☐ Mood board ☐ Freehand thumbnail

4 Key terms: write the letter of the matching description next to each term.

1. **Sectional view**	____	**A** Quick and free early on; no training or investment.
2. **Orthographic drawing**	____	**B** Shows how components fit together.
3. **Exploded view**	____	**C** Gives the manufacturer all views, dimensions and technical details needed.
4. **Working drawing**	____	**D** Shows the interior detail of a product.
5. **Advantage of manual sketching**	____	**E** Shows dimensions and proportions in 2D views (elevation, end elevation, plan).
6. **Advantage of CAD graphics**	____	**F** Easy to edit; many drawings from one model; links to CAM.

5 Describe how graphics can be used to communicate during the design process. (4 marks)

6 Discuss the graphic techniques used at different stages of the design process. You should give clear examples. (8 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?